

PYQBASE — Product Requirements Document (PRD)

Version: 3.0 (Final — Fully Detailed Edition)

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Status:  APPROVED — All pre-development gates passed. Ready for Architecture & Development.

How to read this document: This is the definitive, frozen PRD. It contains the full depth of requirements (merging all details from v1 and v2) alongside the final architectural decisions derived from the Design Review. Where a requirement was modified to protect system scale or budget, a "**Why this was changed**" rationale block explains the engineering decision.

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1. Product Vision

"PYQBASE is India's first unified Previous Year Question Intelligence Platform — where every aspirant can search, analyze, and master questions across all major government exams in one place, powered by smart algorithms and AI-pre-computed explanations."

A student should be able to answer any of the following instantly:

- "Show all questions on Fundamental Rights from 2010–2026 across UPSC, CAPF, and MPSC."*
- "Which Geography subtopics are repeatedly asked? What are the hottest topics of the last 3 years?"*
- "How ready am I for CAPF if I've been preparing for UPSC CSE?"*
- "Show all El Niño questions across UPSC, MPSC, CDS, and CAPF."*

PYQBASE transforms a static PDF archive into a living, intelligent study companion that adapts to every aspirant's journey.

2. Problem Statement

The Core Gap

India has over **50 million competitive exam aspirants** preparing for government jobs. Their most trusted study resource — Previous Year Questions (PYQs) — is scattered, siloed, and inaccessible in any intelligent format.

What Exists Today

Available	Gap
Official PYQ PDFs on government websites	Not searchable, not tagged, not structured
Topic-wise PYQs for individual exams	No cross-exam intelligence
Coaching institute compilations	Expensive, incomplete, no analytics
Standalone test engines	Track right/wrong only — no insight

The Opportunity

No platform has built a **cross-exam, topic-intelligent, analytics-driven PYQ engine**. PYQBASE fills this exact gap.

3. Target Audience

MVP Scope (Phase 1 — Launch)

- **Exams Covered:** UPSC CSE, CAPF (AC), MPSC CSE
- **Geography:** India (pan-India, Hindi & English medium aspirants)
- **PYQ Coverage:** 2021–2025 at launch, expanding backwards as data is available

Phased Expansion

Phase	Exams Added
Phase 1 (MVP)	UPSC CSE, CAPF, MPSC CSE
Phase 2	All Central Exams — NDA, CDS, CISF, SSC CGL, IBPS PO

Demographic Profile

- **Age:** 20–32 years
 - **Device:** Mobile-first (60–70%), Desktop (30–40%)
 - **Language:** English medium (MVP), Hindi medium (future)
 - **Income:** ₹0 to ₹25,000/month (students, fresh graduates)
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4. User Personas

- **Persona 1: Arjun — The Dedicated UPSC Aspirant:** Needs daily revision queue, trend analysis, cross-exam search. Willing to pay ₹299–499 / 3 months.
 - **Persona 2: Priya — The Multi-Exam Strategist:** Needs Cross-Exam Syllabus Overlap Matrix, topic heatmap, MPSC vs UPSC question comparison.
 - **Persona 3: Rohit — The Last-Mile Crammer:** Needs frequency analytics, topic heatmap, mock test generator with CAPF-weighted questions, weak area analysis.
 - **Persona 4: Divya — The Casual Explorer:** Uses free tier for search, heatmap, trend analysis.
 - **Persona 5: Admin — The Content Manager:** Solo founder managing content, tagging, quality control.
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5. Business Goals

6-Month Goals (Post-Launch)

- **Registered Users:** 15,000–30,000
- **Monthly Active Users:** 3,000–6,000
- **Premium Subscribers:** 100–300
- **Questions in DB:** 25,000–40,000 tagged PYQs

Strategic Goals

1. Become the #1 Google result for "UPSC [topic] PYQ" queries.
 2. Build a daily habit loop via SRS queues.
 3. Validate premium conversion before scaling monetization.
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6. Legal & Compliance Requirements

- **LC-01:** Data Protection Notice must be published at /privacy before data is collected.
 - **LC-02:** Explicit, informed consent at signup (no pre-ticked checkboxes).
 - **LC-03:** Self-service Account Deletion available. Implemented as "Soft Delete" (see FR-01.4).
 - **LC-04:** Data Export feature allowing users to download all personal data (JSON/CSV).
 - **LC-05:** 72-hour breach notification protocol.
 - **LC-06:** GST registration and Razorpay automated invoicing.
 - **LC-07:** Public Refund Policy (7-day window) and Terms of Service.
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7. Content Policy & Copyright Framework

- **CP-01:** All question stems and option text must be stored as paraphrased educational adaptations, not verbatim copies.
 - **CP-02:** Attribution notice required on all questions: *"Based on [Exam Name] [Year] examination. Content is an educational adaptation for study purposes."*
 - **CP-03:** PIB RSS is the sole official source for news content; absolutely no scraping of The Hindu or private publishers.
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8. Pre-Development Validation Gates

All pre-development technical blockers were tested and **PASSED** on July 5, 2026:

1. **Gate 1 (OCR):** Tesseract successfully parsed UPSC/CAPF markdown files with near 100% confidence.
 2. **Gate 2 (AI Explanations):** Groq API (Llama-3.3-70B) achieved 100% structured JSON success with ~1000ms latency.
 3. **Gate 3 (Search Benchmark):** Pure Python rank - bm25 achieved <45ms p95 latency at 40,000 documents. We will use rank - bm25 for MVP Search; Typesense/Postgres FTS are not required yet.
 4. **Gate 4 (Paraphrasing):** Groq API successfully rewrote questions while maintaining factual integrity (lexical similarity scores < 60%).
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9. Functional Requirements

FR-01: User Authentication & Accounts

- FR-01.1: Registration via Email/Password or Google OAuth (using Supabase Auth).
- FR-01.2: Free tier users get a 7-day premium trial automatically at signup.
- FR-01.3: Email verification required before saving progress.
- **FR-01.4:** Account deletion implements "soft delete". User account marked with `deleted_at`. If user logs in within 30 days, they see a "Reactivate" button. After 30 days, a cron job permanently deletes data.

Why this was changed (Architecture Review): Immediate hard-deletion cascades through related historical analytics tables, breaking aggregate stats. A 30-day soft-delete is legally compliant with DPDP grace periods while protecting database integrity.

- **FR-01.5:** Auth rate limiting relies entirely on Supabase's native built-in limits.

Why this was changed (Architecture Review): Building custom IP proxying in FastAPI to rate-limit Supabase Auth is an anti-pattern. We leverage the managed service limits directly.

FR-02: Cross-Exam PYQ Search

- FR-02.1: Global search querying across question text, topic, subtopic.
- FR-02.2: Results filterable by Exam, Year, Subject, Topic, Difficulty.
- **FR-02.3:** Pagination uses **Offset-Based Pagination** (Page 1, 2, 3).

Why this was changed (Architecture Review): The original PRD mandated cursor-based pagination. However, because we selected rank - bm25 which returns results sorted by relevance scores rather than sequential IDs, cursor pagination is mathematically complex. Offset pagination is the standard for ranked search engines.

FR-03: Topic Heat Map & Analytics

- FR-03.1: Visual heatmap showing question frequency by topic, served from pre-computed materialized views to ensure <1.5s load times.

FR-04: Trend Analysis & Prediction

- FR-04.1: Weighted Recency Score (2024–2026 gets 3x weight).
- FR-04.2: Statistical "AI Prediction" labels for hot subtopics.

FR-05: PYQ Dashboard

- FR-05.1: Publicly accessible per-topic dashboards showing year-wise distribution and related subtopics.

FR-06 & 07: Cross-Exam Intelligence

- FR-06.1: Exam Similarity Engine showing "Similar Questions" (pre-computed ID arrays).
- FR-07.1: Syllabus Overlap Matrix calculating theoretical and personal readiness for secondary exams.

FR-08: PYQ Revision Mode (Quiz Engine)

- **FR-08.1: Free Tier Limit:** Free users are limited to **30 question ATTEMPTS per day**. Attempting the same question twice in a day counts as 2 attempts.

Why this was changed (Architecture Review): Clarification that the limit tracks rows created in the `user_attempts` table (attempts) rather than tracking a complex set of unique question IDs.

- FR-08.2: Option Elimination Trainer checkboxes with Accuracy Reports (unlocks after 200 questions).
- FR-08.3: Session state is preserved for 24 hours if browser is closed mid-quiz.

FR-09: Spaced Repetition System (SRS)

- FR-09.1: SM-2 algorithm calculates `next_review_date`.
- **FR-09.2:** Database stores `next_review_date` as a PostgreSQL DATE (YYYY-MM-DD) anchored in IST (Indian Standard Time). No UTC conversions needed.

Why this was changed (Architecture Review): The target audience is 100% Indian. Storing as IST DATE strings avoids expensive UTC timezone casting calculations every time the daily queue is loaded, drastically improving database performance.

- FR-09.3: Free tier includes unlimited daily SRS revisions.

FR-10: Smart Mock Tests

- FR-10.1: Free users can generate 1 custom mock test per week. Premium gets unlimited + auto-weak-area generation.

FR-11: Syllabus Coverage

- **FR-11.1:** NLP mapping done via `sentence-transformers` + FAISS.

Why this was changed (Architecture Review): This heavy AI batch job will run locally on the founder's machine once a month to avoid expensive cloud RAM costs. `sentence-transformers` requires 1-2GB of RAM. The resulting `.faiss` index is simply uploaded to cloud storage.

FR-12: AI Explanation System

- FR-12.1: 4-section explanations pre-computed via Groq API.
- FR-12.2: Mandatory "AI-Generated · Verify from official sources" disclaimer on all explanations.

FR-13: Dynamic ELO Difficulty

- **FR-13.1:** Difficulty updates are written to a **Redis queue** per-attempt. A background worker batch-flushes these updates to the PostgreSQL database every 5 minutes.

Why this was changed (Architecture Review): The original PRD required real-time DB updates per attempt. If 1,000 users take a quiz concurrently, updating ELO directly in Postgres triggers massive row-level write contention and locking. Batching via Redis eliminates this critical scaling bottleneck.

- **FR-13.2:** The difficulty label is cached in Redis with a 5-minute TTL. Users see difficulty scores delayed by up to 5 minutes to protect database load.

Why this was changed (Architecture Review): Real-time fetches on every question load will exhaust connection pools. A 5-minute stale read is perfectly acceptable for ELO scores in an educational app.

FR-14: News-to-Question Bridge

- **FR-14.1:** MVP implementation is a **100% manual CRUD form** in the admin panel. Admin types the topic, searches for questions, and publishes.

Why this was changed (Architecture Review): Removed ambiguous references to automated RSS scraping for MVP. Postmortem analysis proved scraping was high-risk and low-ROI. Manual curation guarantees quality control for the Alpha launch.

FR-15: User Progress Dashboard

- FR-15.1: Streak calendar based on IST day boundaries.

FR-16: Admin Panel

- FR-16.1: Secure content management (CRUD for questions, explanations, news topics).
- **FR-16.2:** Admin access requires a specific Supabase Auth account with the `is_admin` custom JWT claim.

- **FR-16.3:** To prevent **Admin Bundle Exposure**, the admin panel must be built as a separate Next.js entrypoint (or separate deployment) so that admin API routes and components are never shipped in the public client-side JavaScript bundle.

Why this was changed (Architecture Review): Bundling admin views behind a simple React router check exposes internal API endpoints to malicious users inspecting the JS source code. Complete isolation secures the attack surface.

- FR-16.4: Immutable audit trail for all content modifications.

FR-17: Email Notifications

- FR-17.1: Daily SRS reminder emails sent at 7:00 AM IST.
- **FR-17.2:** Integration implemented using **Resend API**.

Why this was changed (Architecture Review): Solidified the SDK choice to unblock development. Resend offers the best Python/Next.js tooling for MVP scale.

FR-18 & 19: Onboarding & Error Triage

- FR-18.1: 3-step onboarding with mandatory 15-question starter session to populate Day 2 SRS.
 - FR-19.1: Dedicated P1-P4 triage workflows for user error reports.
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10. Non-Functional Requirements

- **Performance:** Page load time (LCP) under 2 seconds. Search results return in under 500ms (p95). Quiz engine renders next question in under 200ms.
 - **Reliability:** Platform must maintain 99.5% uptime.
 - **Caching:** Redis instance alongside FastAPI for heatmaps, ELO, and similarity arrays.
 - **Database Pooling:** PgBouncer enforces max 20 server connections to PostgreSQL to prevent "too many connections" errors.
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11. User Stories

Student Users

- **As Arjun,** I want to see which subtopics of Geography are asked most frequently, so I can prioritize my revision.
- **As Arjun,** I want to log in every morning and see exactly which questions I need to review today (SRS), so I don't forget past mistakes.
- **As Priya,** I want to see how much of the MPSC syllabus is covered by my UPSC preparation, so I know if I can pass MPSC with minimal extra study.

- **As Rohit**, I want to generate a 50-question mock test specifically targeting my weak areas in CAPF Polity, so I can maximize my ROI in the final 3 months.
- **As Divya**, I want to search for "El Nino" without logging in, so I can see if the platform has the questions I'm looking for before I commit to an account.

Admin Users

- **As Admin**, I want to import a parsed Markdown file of a question paper and have the system auto-extract questions, options, and answer keys.
- **As Admin**, I want to review AI-generated explanations before they go live for high-frequency questions.
- **As Admin**, I want to see a queue of user-reported errors (P1 for wrong answer keys) and resolve them instantly to protect platform trust.

12. Acceptance Criteria

- **Search Engine:** User searches "Article 14". System returns all related questions across UPSC, CAPF, MPSC within 500ms. The UI shows matched keywords highlighted.
- **SRS Logic:** User gets a question wrong. The system schedules it for review tomorrow. User gets it right tomorrow. System schedules it for review 3 days later.
- **Heatmap Engine:** User clicks "History". System renders a color-coded heatmap of "Ancient", "Medieval", "Modern" within 1.5 seconds, even if the database has 100,000 rows.
- **Mock Test Generator:** User requests 20 questions from "Geography" with "Hard" difficulty. System returns a test with negative marking configured. At submission, an analytics report compares time-per-question.

13. Freemium Boundary Definition

Feature	Free Tier	Premium Tier
Cross-Exam Search	Unlimited	Identical
Topic Heatmap & Analytics	Unlimited	Identical
PYQ Quiz Engine	30 attempts per day (resets midnight IST)	Unlimited
Elimination Trainer	Available within free quota	Available
SRS / Daily Revision Queue	Unlimited (SRS queue is never gated)	Identical
Smart Mock Test (custom)	1 per week (max 25 questions)	Unlimited (max 100 questions)
Smart Mock Test (weak-area)	Not available	Available
Mock Test History	Last 3 tests only	Unlimited history

Feature	Free Tier	Premium Tier
Syllabus Coverage Heatmap	Free (Global view)	Identical
Personal Syllabus Coverage	Not available	Available

Paywall UX Principle: Every premium gate must show value first. Never hard-block without showing what the user *would* get. Gates are modal-based so the user stays in context.

14. Feature List

#	Feature	Category	Free / Premium
F-01	Global Cross-Exam PYQ Search	Core	Free
F-02	Topic Heat Map	Analytics	Free
F-03	Frequency Analytics (charts)	Analytics	Free
F-04	Trend Analysis & Prediction Engine	Analytics	Free
F-05	PYQ Dashboard (per topic)	Analytics	Free
F-06	Cross-Exam Syllabus Overlap Matrix	Intelligence	Free
F-07	Exam Similarity Engine	Intelligence	Free
F-08	PYQ Revision Mode	Practice	Free (30/day)
F-09	Option Elimination Trainer	Practice	Free
F-10	Elimination Accuracy Report	Analytics	Free (after 200 Qs)
F-11	AI Explanations	Learning	Free
F-12	Spaced Repetition System (SM-2)	Practice	Free
F-13	Dynamic ELO Difficulty Rating	Intelligence	Free
F-14	Smart Mock Test Generator	Practice	Free (limited) / Premium
F-15	Personal Syllabus Coverage tracking	Strategy	Premium
F-16	News-to-Question Bridge	Daily Hook	Free

15. MVP Features (Alpha Launch & Public Launch)

Alpha Launch (Invite-Only — 50–100 users)

1. **User Auth:** Supabase (Email + Google), free tier limits, email verification.
2. **User Onboarding Flow:** Exam selection, 3-tooltip tour, 15-question starter session.
3. **Cross-Exam PYQ Search:** Offset pagination, filters.
4. **Topic Heat Map:** Served from materialized views.
5. **PYQ Revision Mode:** 30/day free limit, AI Explanations.
6. **SRS:** SM-2, Daily Revision Queue, IST anchor.
7. **Admin Panel:** Content management, news bridge management.

Public Launch (Added on top of Alpha)

1. **Trend Analysis:** Weighted recency score, Prediction labels.
 2. **Exam Similarity Engine:** Pre-computed similar question arrays.
 3. **Cross-Exam Syllabus Overlap Matrix.**
 4. **Smart Mock Test Generator:** Custom (1/week free) + weak-area (premium).
 5. **Syllabus Coverage Mapper:** Heatmap + never-tested zones.
 6. **Dynamic ELO Difficulty Rating:** Redis-cached.
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16. Data Pipeline Specification & Schema

Question Record Schema (JSON format):

```
{
  "exam": "UPSC_CSE",
  "year": 2024,
  "paper": "Prelims_GS1",
  "question_number": 47,
  "question_stem": {"en": "Paraphrased text..."},
  "options": {
    "A": {"en": "Option A..."},
    "B": {"en": "Option B..."}
  },
  "correct_option": "B",
  "question_type": "direct_mcq",
  "has_image": false,
  "image_url": "https://supabase.../images/q47.jpg",
  "image_description": "Map description...",
  "parse_confidence": 0.97,
  "subject": "Geography",
  "topic": "Climatology",
  "subtopic": "Monsoon",
  "syllabus_point_id": 412
}
```

Why this was changed (Architecture Review): Added `image_url` to explicitly

define how diagrams/maps are loaded by the frontend. Modified text fields to use `{"en": "..."} JSONB structure to allow Phase 2 Hindi addition without massive schema rewrites.`

Taxonomy Architecture: The system uses strict relational tables (`subjects` → `topics` → `subtopics`) linked by foreign keys, NOT free-text strings.

Why this was changed (Architecture Review): The Overlap Matrix relies on rigid hierarchical structures. Free-text strings invite typos that break algorithmic overlap logic.

17. Content Quality Standards

- No question is published without matching answer key, subject categorization, and verified parsing (confidence ≥ 0.90 or manual review).
 - Tier 1 explanations (high frequency) are 100% human-reviewed before launch.
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18. Future Features & Architecture Hooks

Phase 2: Exam & Language Expansion

- Add NDA, CDS, CISF, SSC CGL.
- The JSONB schema natively supports `{"hi": "text"}` addition.

Phase 5: B2B / Institutional Licensing

- **Architecture Rule:** The MVP database is **strictly single-tenant (B2C)**. Do not implement `institute_id` or multi-tenant RLS policies at this stage.

Why this was changed (Architecture Review): B2B is a Year 2 aspiration. Over-engineering the database schema for multi-tenancy on Day 1 drastically slows down MVP development speed.

19. Monetization

- **Pricing:** ₹299 (Monthly), ₹799 (3-Month), ₹2,499 (Annual).
 - **Break-even:** The financial model confirms break-even is achieved at just 25 active subscribers (to cover Vercel, Railway, Supabase Pro, and Resend costs).
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20. Risks

R-01: Data Quality Risk ☐ CRITICAL

- **Risk:** OCR pipeline produces incorrect question stems or answer keys leading to widespread trust damage.
- **Mitigation:** Parse confidence scoring (hold < 0.90 for admin review). Alpha launch strictly capped at 3,000 highly curated questions.

R-02: Copyright / Legal Risk ☐ CRITICAL

- **Risk:** UPSC or state PSC bodies issue legal notice claiming ownership over examination content.
- **Mitigation:** Paraphrase-based content model — no verbatim reproduction (Validated by Gate 4). Clear educational-use disclaimers. Never provide bulk-download of PDF question sets.

R-03: DPDP Act Non-Compliance ☐ CRITICAL

- **Risk:** Platform collects user data without proper consent or erasure mechanisms.
- **Mitigation:** Full compliance framework implemented (LC-01 to LC-06).

R-04: Solo Founder Bandwidth Risk ● MEDIUM

- **Risk:** One person managing data pipeline, development, content, and support leads to burnout and slow shipping.
- **Mitigation:** Alpha launch scope reduced to 3,000 questions. Manual news bridge in MVP eliminates fragile RSS pipeline maintenance overhead.

R-05: User Retention at Cold Start ● MEDIUM

- **Risk:** Users visit once, see empty SRS queue, don't return.
- **Mitigation:** Onboarding flow (FR-18) with a mandatory starter session ensures Day 2 SRS queue is non-empty. Daily SRS reminder emails sent at 7:00 AM IST.

21. Operational Requirements

Adding new exams (e.g., NDA) in the future is achievable via the admin panel (data entry, syllabus upload, run pipeline) and must not require a code deployment.

22. Infrastructure & DevOps Requirements

- **Frontend:** Next.js hosted on Vercel.
- **Backend:** FastAPI + Redis hosted on Railway.

- **Database:** PostgreSQL hosted on Supabase Pro.
 - **CI/CD:** GitHub Actions (Lint, Test, Deploy to Staging, Manual push to Prod).
 - **Secrets:** Environment variables only; no secrets in Git.
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23. Success Metrics

- **Day-2 Retention:** $\geq 35\%$
 - **Alpha Target:** 3,000 perfectly curated UPSC CSE questions.
 - **Public Launch Target:** 15,000 questions (UPSC + CAPF + MPSC).
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24. Roadmap

- **Phase 0 & 0.5 (Completed):** Pre-architecture gates passed. Architecture decisions locked.
 - **Phase 1 (Current):** Alpha Development Setup (Next.js, FastAPI, Supabase).
 - **Phase 1.5:** Public Launch (Weeks 11-14).
 - **Phase 2 & 3:** Growth, App Development, Gamification.
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25. Tech Stack Summary

Layer	Technology
Frontend	Next.js (React, Tailwind)
Backend	FastAPI (Python, Async)
Database	PostgreSQL (Supabase Pro) + PgBouncer
Cache	Redis
Auth	Supabase Auth (Email + Google)
Search	rank - bm25 (Python)
AI Explanations	Groq API (Llama-3.3-70B)
Semantic Mapping	sentence-transformers + FAISS (Run locally by Founder)
Email	Resend API
